Polarise Telecom Technology

Comprehensive Project Architecture, Telephony Integration & Operational Report

Client: Polaris Educational Consultancy	**Version:** 1.0.0 Production
System: Admissions Telecalling & PBX CRM	**Date:** October 2026 | Deployment: Vercel Live

1. Executive Summary

Polarise Telecom Technology is an enterprise-grade Cloud Telecalling and Student/Parent Admissions CRM developed specifically for **Polaris Educational Consultancy**. The platform bridges high-volume educational counseling workflows with cloud-based telecom carrier trunks (Exotel PBX). It enables educational counselors to conduct validated outbound calling, manage lead pipelines (100–200 contacts per counselor), record admissions conversations, receive forwarded parent inquiries, and deliver live supervisor telemetry—all while preserving complete privacy and data confidentiality.

2. Telephony Configuration (TelephonyConfig) & Its Purpose

The **TelephonyConfig** module serves as the mission-critical abstraction layer between the CRM application and telecom voice carrier backbones. Without telephony configuration, a CRM is merely a static contact database; with TelephonyConfig, the system acts as an interactive, carrier-integrated virtual call center.

Virtual Number Masking (Caller ID)	All outbound calls made by counselors display the verified Polaris virtual number (+918047190000). Counselors' personal mobile numbers remain 100% confidential, while parents see a trusted, professional institutional caller ID.
Click-to-Call PBX Bridging	When an employee initiates a call from the dashboard, the cloud PBX first rings the counselor's registered extension/phone. Upon answer, it instantaneously dials the student/parent and bridges both audio streams in real-time.
Dual Operating Mode Engine	**Demo Sandbox Mode:** Provides realistic call simulations with synthetic WAV audio waveform generation for training and presentations with zero carrier bills. **Live Exotel Mode:** Connects directly to Exotel REST APIs using Account SID, API Key, API Token, and Subdomain.
Automated Call Recording & Storage	Every call initiated through the PBX is automatically captured from the moment of connection until disconnect. Audio files are indexed and made available strictly to administrative personnel.
Idempotent Webhook Callbacks	Listens for real-time carrier events on /api/telephony/webhooks/exotel. Automatically updates call duration, answer timestamp, disposition, and recording SID without continuous client polling.
Smart Call Forwarding Routing	Empowers counselors to toggle call forwarding, automatically routing inbound parent callbacks to mobile lines or designated fallback counseling desks.

3. System Architecture & Technology Stack

Component	Technologies Used	Functional Responsibility
Frontend SPA	React 18, TypeScript, Vite, Tailwind CSS, Lucide Icons	Chroma Waves dynamic canvas, counselor calling desk, admin live supervision, analytics graphs
Backend API	Python 3.10+, FastAPI, SQLAlchemy, Pydantic, Uvicorn	Async REST endpoints, JWT role security, OpenPyXL parser, carrier webhook handlers
Database	SQLite (with /tmp sync for serverless)	Zero-cold-start relational persistence for 13 staff accounts, calls, recordings, and lead queues
Cloud Telephony	Exotel Voice PBX & Simulated Wave Engine	PSTN/GSM voice bridging, virtual caller ID masking, automated audio recording
Deployment	Vercel Serverless Edge Runtime	Full-stack monorepo deployment with automated CI/CD and public SSL access

4. User Personas & Permissions Matrix

Access control is strictly governed by Role-Based Access Control (RBAC). A total of 13 pre-configured staff accounts are pre-seeded in the database to accommodate team operations:

Role Level	Default Accounts	Call Recordings	Lead Allocation	PBX Config
Super Admin	Dr. Arvind Menon (admin1@polarisedu.in)	FULL Access (Listen & Export)	All Employees	Full Access
Operations Admin	Pooja Sharma (admin2@polarisedu.in)	FULL Access (Supervisory)	All Employees	Read Only
Reporting Admin	Karthik Iyer (admin3@polarisedu.in)	FULL Access (Analytics)	All Employees	Read Only
Counselor (10 Staff)	Rohan, Sneha, Ananya, Vikas, Meera, Aditya, Priya, Rahul, Divya, Farhan	**STRICTLY HIDDEN**	Assigned Leads Only	Hidden

5. Step-by-Step Operational Workflows

Step 1: Security & Locked Profile Protection

Users authenticate via the login portal with the Chroma Waves backdrop. If an administrator locks an employee's profile from the Live Status board, the system revokes access immediately. Any subsequent login attempts present a bold security alert advising them to 'Contact Administrator'.

Step 2: Excel / CSV Bulk Lead Ingestion & Distribution

The administrator uploads bulk student spreadsheets (.xlsx / .csv). The custom ingestion engine strips out sensitive cost/pricing columns, validates phone numbers against the 10-digit standard, and creates student lead profiles. The admin can then allocate 100 to 200 leads to each of the 10 counselors either through one-click balanced distribution or individual selection.

Step 3: Admissions Counselor Calling Desk

Counselors see only their designated leads. Clicking any student opens the calling interface. Built-in phone validation prevents accidental dialing if a number contains more or less than 10 digits. When a call completes, counselors select a mandatory call disposition (e.g. Interested, Callback Requested, Admission Confirmed, Not Reachable) and enter counseling notes. Recordings are hidden from the employee.

Step 4: Real-time Live Agent Supervision

Administrators monitor active team availability on the Live Status Board, indicating the exact count and identity of counselors currently Available, On Call, in Wrap-up, or Offline, ensuring balanced workload and active phone coverage.

Step 5: Call Recordings Audio Station

Only administrators possess access to the audio vault. The built-in HTML5 player supports scrubber seek, volume control, and duration markers, allowing directors to evaluate counselor pitch quality and student interactions.

Step 6: Day-wise, Weekly & Monthly Analytics Reports

Dynamic calculation computes call metrics directly from verified call logs. The UI presents multi-colored grouped bar charts (each day highlighted in vibrant gradients) and a chromatic spline trend graph. Clicking any period reveals the telemetry side drawer with granular call detail records.

Step 7: Corporate Data Export Suite

Administrators can filter and generate 5 standardized reports (Call Detail Records, Staff Performance, Admissions Conversion, Inbound Routing, Audit Trails) with live column schema previews and one-click Excel (.xlsx) / CSV downloads.

6. Production Deployment & Live Infrastructure

Production Web URL:	https://polarise-telecom.vercel.app
GitHub Repository:	https://github.com/charan5177/polarise-telecom
Vercel Project Scope:	shaikasif346s-projects / polarise-telecom
Backend Health Check:	https://polarise-telecom.vercel.app/api/health (200 OK)
Default Master Credentials:	admin1@polarisedu.in | Polaris@2026

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